

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase III, Multicenter, Randomized, Placebo Controlled, Double-blind Study to Assess Efficacy and Safety of Crizanlizumab (5 mg/kg) Versus Placebo, With or Without Hydroxyurea/Hydroxycarbamide Therapy, in Adolescent and Adult Sickle Cell Disease Patients With Frequent Vaso-Occlusive Crises
Short Title:	A Study to Investigate the Efficacy and Safety of Crizanlizumab (5 mg/kg) Compared With Placebo in Adolescent and Adult Sickle Cell Disease Patients Who Experience Frequent Vaso-Occlusive Crises (SPARKLE)
Protocol Number:	1012821946284250696
NCT Number:	NCT06439082
Sponsor Name:	Novartis
Investigational Product:	crizanlizumab
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Annualized rate of VOCs that are healthcare professional (HCP)-managed (including VOCs leading to management at a health care facility or those managed via remote consultation) in each treatment arm: Vaso occlusive crisis (VOC) is defined as a pain crisis (acute onset of pain for which there is no other medically determined explanation other than vaso-occlusion) lasting for at least 4 hours which is treated as per local guidelines with standard of care therapy used to treat VOC. Acute chest syndrome (ACS), priapism and hepatic or splenic sequestration will be considered VOC in this study. VOCs included are those HCP-managed in a healthcare facility and HCP-managed via remote consultation. Annualized rate of VOC events = (Number of VOC events * 365)/(number of days in the observation period). Observation period = time from date of randomization to minimum of (last dose date until treatment discontinuation + 27 days, date of initiation or discontinuation of HU/HC or L-Glutamine (or other therapies such as Voxelotor and erythropoietin therapies to treat SCD and/or to prevent/reduce VOCs), date of randomization + 365 days).
Secondary Obj.:	1. The annualized rate of all VOCs including VOCs that are HCP-managed (either at a health care facility or via remote consultation) as well as those that are self-managed

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

without recommendations from HCP during the event in each treatment arm 2. Annualized rate of VOC by subtype of management in each treatment arm over the planned 52-week period. 3. The time to first VOC that is HCP-managed (including VOCs leading to management at a health care facility or those managed via remote consultation) between treatment arms.

2.2 Background & Rationale

A phase III, multi-center, randomized, placebo-controlled, double-blind study to assess efficacy and safety of crizanlizumab (5 mg/kg) versus placebo, with or without hydroxyurea/hydroxycarbamide therapy, in adolescent and adult Sickle Cell Disease patients with frequent vaso-occlusive crises.

3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE3, Status: RECRUITING

Target N: 315

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- Participants must be aged 12 years and older on the day of signing informed consent. Adolescents include participants aged 12 to <18 years old and adults include participants aged 18 years and older.
- Confirmed diagnosis of SCD by Hb electrophoresis or high-performance liquid chromatography (HPLC) (performed locally or by central laboratory if not available locally). All SCD genotypes are eligible.
- Experienced 4 to 12 VOCs (refer to Section 8.3.1 for study definition of VOC) that are HCP-managed (including VOCs leading to management at a health care facility or those managed via remote consultation) within the 12 months prior to the screening visit. Baseline VOCs are determined by medical history and are required to be documented at source.
- If the participant is on HU/HC, they must be taking it for at least 6 months and at stable dose for at least 3 months prior to the Screening visit and plan to continue taking it at the same dose and schedule until at least the participant has reached 52 weeks of the planned study treatment. Participants who have initiated HU/HC 6-12 months prior to the screening visit must have evidence of insufficient control of acute pain despite initiation. These participants must have a cumulative of 4-12 VOCs in the 12 months prior to the screening period, with at least 2 during the last 6 months while on HU/HC. If receiving erythropoietin stimulating agent, the participant must have been receiving the drug for at least 6 months prior to screening visit and plan to continue taking the drug at the same dose and schedule until the participant has reached 52 weeks of the planned study treatment.

Participants who have not been receiving HU/HC, and/or erythropoietin stimulating agent must not have received it for at least 6 months prior to screening visit.

Key Exclusion Criteria:

- Fewer than 4 or more than 12 VOCs that are HCP-managed (including VOCs leading to management at a health care facility or those managed via remote consultation) within the 12 months prior to screening visit as determined by medical

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

history and documented at source.

2. History of stem cell transplant and/or gene therapy.

3. Received blood products within 30 days prior to Week 1 Day 1 dosing.

4. Any documented history of a clinical stroke or intracranial hemorrhage, or an uninvestigated neurologic finding within the past 12 months before screening visit. Silent infarct only present on imaging is not excluded.

5. Participating in a chronic transfusion program (pre-planned series of transfusions for prophylactic purposes) and/or planning to undergo an exchange transfusion during the duration of the study; episodic transfusion in response to worsened anemia or VOC is permitted.

6. Contraindication or hypersensitivity to any drug or metabolites from similar class as study drug or to any excipients of the study drug formulation. History of severe hypersensitivity reaction to other monoclonal antibodies, which in the opinion of the investigator may pose an increased risk of serious infusion reaction.

4.2 Exclusion Criteria

1. Fewer than 4 or more than 12 VOCs that are HCP-managed (including VOCs leading to management at a health care facility or those managed via remote consultation) within the 12 months prior to screening visit as determined by medical history and documented at source.

2. History of stem cell transplant and/or gene therapy.

3. Received blood products within 30 days prior to Week 1 Day 1 dosing.

4. Any documented history of a clinical stroke or intracranial hemorrhage, or an uninvestigated neurologic finding within the past 12 months before screening visit. Silent infarct only present on imaging is not excluded.

5. Participating in a chronic transfusion program (pre-planned series of transfusions for prophylactic purposes) and/or planning to undergo an exchange transfusion during the duration of the study; episodic transfusion in response to worsened anemia or VOC is permitted.

6. Contraindication or hypersensitivity to any drug or metabolites from similar class as study drug or to any excipients of the study drug formulation. History of severe hypersensitivity reaction to other monoclonal antibodies, which in the opinion of the investigator may pose an increased risk of serious infusion reaction.

11. Study Identifiers & Governance

Primary ID: 1012821946284250696

Registry Number: NCT06439082

Sponsor: Novartis

Study Title: A Study to Investigate the Efficacy and Safety of Crizanlizumab (5 mg/kg) Compared With Placebo in Adolescent and Adult Sickle Cell Disease Patients Who Experience Frequent Vaso-Occlusive Crises (SPARKLE)

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____